Eventually periodic sequence

Tags:

Time Limit: 1 s Memory Limit: 128 MB
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Description

Given is a function

$f: 0..N \dashrightarrow 0..N$

for a non-negative

N

and a non-negative integer

n

$\leq$

N

. One can construct an infinite sequence

$F = f^1(n), f^2(n), \ldots f^k(n) \ldots$

, where

$f^k(n)$

is defined recursively as follows:

$f^1(n) = f(n)$

and

$f^{k+1}(n)$

=

$f(f^k(n))$

.

It is easy to see that each such sequence F is eventually periodic, that is periodic from some point onwards, e.g 1, 2, 7, 5, 4, 6, 5, 4, 6, 5, 4, 6 ... . Given non-negative integer $N \leq 11000000$, $n \leq N$ and f, you are to compute the period of sequence F.

Each line of input contains N, n and the a description of f in postfix notation, also known as Reverse Polish Notation (RPN).

The operands are either unsigned integer constants or *N* or the variable *x*. Only binary operands are allowed: + (addition), * (multiplication) and % (modulo, i.e. remainder of integer division). Operands and operators are separated by whitespace. The operand % occurs exactly once in a function and it is the last (rightmost, or topmost if you wish) operator and its second operand is always *N* whose value is read from input. The following function:

```
2 x * 7 + N %
```

is the RPN rendition of the more familiar infix

`(2*x+7)%N`

. All input lines are shorter than 100 characters. The last line of input has

N

equal 0 and should not be processed.

For each line of input, output one line with one integer number, the period of *F* corresponding to the data given in the input line.

Samples

input Copy

```
10 1 x N %
11 1 x x 1 + * N %
1728 1 x x 1 + * x 2 + * N %
1728 1 x x 1 + x 2 + * * N %
100003 1 x x 123 + * x 12345 + * N %
0 0 0 N %
```

output Copy

```
1
3
6
6
369
```

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